MATERIAL SCIENCE · SEMESTER 1 · UNIT 3

Unit 3 Review: Metals

ANSWER KEY

Show your work for calculations and use full sentences for explanations. Answers are in the answer key that comes with this worksheet in Classboard.

A · BONDING AND STRUCTURE

1. Explain why copper wire can be bent into a loop but a ceramic mug shatters when dropped. Use the word "bond" in your answer.

   Metallic bonds are a shared sea of electrons that doesn't point in fixed directions, so layers of copper atoms can slide and re-bond. A ceramic's ionic/covalent bonds are fixed, so shifting layers push apart and the mug cracks.

2. Match each metal to its usual crystal structure (BCC, FCC or HCP): iron at room temperature, aluminum, zinc.

   Iron: BCC · Aluminum: FCC · Zinc: HCP

3. Sample X has large grains and sample Y has tiny grains. Same metal. Which is stronger, and why?

   Y. More grain boundaries block dislocations from moving, so it resists permanent bending.

B · STRESS AND STRAIN

4. A bar with a cross-section of 60 mm² carries 12,000 N. Find the stress in MPa.

   12,000 N ÷ 60 mm² = 200 N/mm² = 200 MPa

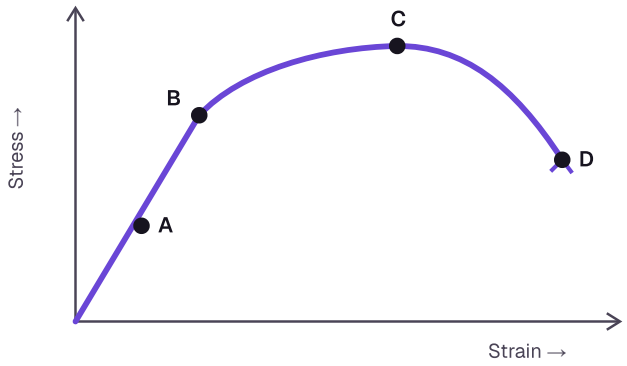
5. The same bar was 150.0 mm long and stretched to 150.3 mm (still elastic). Find the strain.

   0.3 ÷ 150.0 = 0.002

6. Use your answers to 4 and 5 to find Young's modulus.

   200 MPa ÷ 0.002 = 100,000 MPa = 100 GPa

7. Label points A–D on the curve: yield point, fracture, ultimate tensile strength, a point in the elastic region.



A: elastic region · B: yield point · C: ultimate tensile strength · D: fracture

C · HEAT TREATMENT AND WORK HARDENING

8. Three bobby pins are heated red-hot. Pin 1 cools slowly in air. Pin 2 is plunged into water. Pin 3 is plunged into water, then reheated gently and cooled. Predict what each does when you try to bend it.

   Pin 1 (annealed): bends easily and stays bent. Pin 2 (quenched): hard and snaps. Pin 3 (tempered): springy; bends and springs back without snapping.

9. Why does a paper clip break after you bend it back and forth many times?

   Work hardening: each bend adds and tangles dislocations, so the bent spot gets harder and more brittle until it cracks.

D · ALLOYS AND MELTING

10. Name what each alloy is made of: brass, bronze, steel, sterling silver.

    Brass: copper + zinc · Bronze: copper + tin · Steel: iron + carbon · Sterling silver: 92.5% silver + copper

11. Is steel a substitutional or interstitial alloy? Explain.

    Interstitial: the small carbon atoms fit in the gaps between iron atoms.

12. Tin melts at 232 °C and lead at 327 °C. Why do electronics use 63% tin / 37% lead solder (melting point 183 °C) instead of either pure metal? What is this mix called?

    It's the eutectic: it melts all at once at the lowest temperature of any tin-lead mix, so it can join parts without overheating them.

E · MAKING AND PROTECTING METALS

13. Number these lost-wax casting steps in order: __ pour molten metal · __ carve the wax model · __ burn out the wax · __ break off the plaster · __ cover the model in plaster

    4 · 1 · 3 · 5 · 2 (carve the wax → cover it in plaster → burn out the wax → pour the metal → break off the plaster)

14. Give two ways to keep a steel railing outside from rusting, and explain how one of them works.

    Any two: paint/coat it, galvanize it (zinc coating), use stainless steel. Example: galvanizing puts zinc on the surface, which corrodes first and protects the steel.

F · CLAIM, EVIDENCE, REASONING

15. A blacksmith quenches a knife blade and then tempers it. Write a claim about why both steps are needed, give evidence (from the bobby pin lab or the stress-strain curve), and explain your reasoning.

    Claim: quenching makes the blade hard enough to hold an edge, and tempering makes it tough enough not to snap. Evidence: the quenched pin snapped while the tempered pin bent and sprang back. Reasoning: quenching traps a hard, brittle structure; tempering relaxes some of it, trading a little hardness for toughness.